

Anchoring Effects in Large Language Models: A Controlled Benchmark and Mitigation Study

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Paper under double-blind review

Abstract

1 Anchoring bias—the tendency for estimates to gravitate toward contextually
2 salient but irrelevant numbers—is well documented in human cognition
3 yet poorly characterized in large language models (LLMs). We introduce
4 a controlled benchmark of 108 numeric-estimation items spanning
5 three anchor-delivery mechanisms (prompt-embedded anchors, in-context-
6 learning demonstrations, and retrieval-augmented / tool-call contexts), evaluated
7 across six LLMs from three model families (Llama 3 1B/3B/8B,
8 GPT-4o-mini, Claude 3.5 Haiku). Using a Normalized Anchoring Index
9 (nai), we find that all models exhibit statistically significant anchoring
10 on prompt-embedded anchors (nai up to 0.41), with the effect modulated
11 by model scale and anchor-delivery channel. We evaluate nine mitigation
12 strategies—three prompt-level, two structural, two post-hoc, and a novel
13 logit-level Anchor-Aware Logit Correction (AALC)—finding that AALC at
14 $\alpha=1.0$ reduces external-suite nai to near zero for open-weight models while
15 preserving output quality. [TODO: Add final sentence on next steps once
16 mitigation results are finalized.]

17 1 Introduction

18 When asked to estimate a quantity, humans systematically shift their answers toward arbitrary
19 reference points—a phenomenon known as anchoring bias (Tversky & Kahneman,
20 1974; Kahneman, 2011). As LLMs are increasingly deployed in high-stakes domains such as
21 medical triage, financial risk assessment, and operational planning, understanding whether
22 they inherit this bias is both scientifically interesting and practically important.

23 Recent work has begun to probe cognitive biases in LLMs (Jones & Steinhardt, 2022; ?, ?),
24 but existing studies typically (i) test a single anchor-delivery mechanism (usually prompt-
25 embedded), (ii) evaluate few models, and (iii) lack controlled ground-truth labels that separate
26 anchoring from mere accuracy degradation.

27 We address these gaps with the following contributions:

- 28 1. Controlled benchmark. A 108-item dataset (AnchorBench-v0.2) with five anchor-
29 delivery suites (external, ICL, conversation-history, RAG, tool), three application
30 fields (health, finance, operations), and gold-standard reference answers derived
31 from latent signals.
- 32 2. Multi-model evaluation. Systematic measurement of anchoring across six LLMs
33 spanning three families and three scale tiers (1B–8B open-weight; GPT-4o-mini;
34 Claude 3.5 Haiku), using the Normalized Anchoring Index (nai), Total Variation
35 Distance (tvd), and Wasserstein-1 (W_1) metrics.
- 36 3. Mitigation taxonomy and evaluation. Nine mitigation strategies organized into
37 prompt-level (B1–B3), structural (EFR, CxDP), post-hoc (B4, PLD), and logit-level
38 (AALC) categories, evaluated with bootstrap confidence intervals and Benjamini-
39 Hochberg correction.
- 40 4. Stress tests. Four robustness probes (anchor magnitude, position, format, RAG-
41 snippet order) revealing model-specific vulnerabilities.

This paper reports proof-of-concept (PoC) results. We plan to scale the benchmark to ≥ 500 items, add more models, and extend to multi-turn RAG and tool-use settings.

2 Related work

Cognitive biases in LLMs. Several studies have examined whether LLMs exhibit human-like cognitive biases. Jones & Steinhardt (2022) proposed a taxonomy mapping human heuristics to LLM failure modes. ? found that GPT-family models exhibited anchoring and framing effects that diminished with instruction tuning. ? systematically tested 14 biases across multiple LLMs. [TODO: Add 2-3 more citations on LLM cognitive biases.]

Anchoring in LLMs. ? distinguished between instructed and default anchoring, showing that models differ in susceptibility. The tum-nlp/cognitive-biases-in-llms dataset (TUM-NLP, 2024) provides control/treatment pairs for multiple biases including anchoring, but uses ordinal scales (1-9) rather than continuous numeric estimates. [TODO: Add citations for ICL-based and RAG-based anchoring studies.]

Mitigation. Chain-of-thought prompting (Wei et al., 2022) and self-consistency (Wang et al., 2023) are general strategies that may reduce bias. Contrastive decoding methods such as DoLa (Chuang et al., 2024) operate at the logit level. To our knowledge, no prior work has proposed a logit-level correction specifically targeting numeric anchoring.

3 Dataset design

Item structure. Each item in AnchorBench-v0.2 specifies: (i) a latent signal $\theta \in [15, 85]$, (ii) a gold-standard answer y^* derived from θ with Gaussian noise ($\sigma=12$), (iii) low and high anchor values, (iv) prompts for each experimental condition, and (v) metadata (field, domain, template). The answer space is constrained to integers in $[0, 100]$, representing percentages.

Anchor-delivery suites. We organize items into five suites:

- External (30 items): a prompt-embedded anchoring question (e.g., “is the rate higher or lower than X%?”) appears before the estimation instruction.
- ICL (12 items): the anchor is delivered via 3-shot in-context demonstrations whose answers are biased toward the low or high anchor.
- Conversation history (12 items): the anchor is the model’s own prior answer (turn 1), re-presented in the context of a follow-up question (turn 2).
- RAG (12 items): anchoring numbers appear in retrieved snippets prepended to the prompt.
- Tool (12 items): anchoring numbers appear in simulated tool-call outputs.

Across suites, 30 items belong to the external suite and 78 to extension suites (ICL, conversation history, RAG, tool). Three application fields—health, finance, operations—contribute 6 domains total.

Conditions. Every external-like item has three conditions: control (no anchor), low_anchor (anchor $\leq \theta$), and high_anchor (anchor $\geq \theta$). The anchor gap is 60 percentage points by construction.

4 Experimental setup

Models. We evaluate six models across two inference backends:

- Open-weight (HuggingFace Transformers): Llama-3.2-1B-Instruct, Llama-3.2-3B-Instruct, Llama-3.1-8B-Instruct (Grattafiori et al., 2024).

- API (OpenRouter): GPT-4o-mini (OpenAI, 2024), Claude-3.5-Haiku (Anthropic, 2024), Llama-3.1-8B-Instruct (served via OpenRouter).

Inference settings. Temperature = 0.7, top- p = 1.0, max tokens = 8 (128 for chain-of-thought). Each item-condition pair is sampled 5 times. A system prompt instructs the model: “You are a helpful assistant. Follow the instructions exactly.”

Parsing. We extract the last integer in $[0, 100]$ from each response, excluding anchor echoes detected via regex matching against the prompt text. Parse rates exceed 95% for all models. [TODO: Report exact per-model parse rates.]

5 Metrics

Normalized Anchoring Index (nai). For each item i with low-anchor samples $\{y_j^L\}$ and high-anchor samples $\{y_j^H\}$, we define

$$\text{NAI}_i = \frac{\bar{y}_i^H - \bar{y}_i^L}{60}, \quad (1)$$

where $\bar{y}$ denotes the sample mean and 60 is the anchor gap. $\text{NAI} = 1$ indicates complete susceptibility; $\text{NAI} = 0$ indicates no anchoring.

Distributional metrics. We complement nai with per-item Total Variation Distance (tvd) and Wasserstein-1 distance (W_1) between the low-anchor and high-anchor empirical distributions.

Relative Reduction (rr). For a mitigation m applied to model k , we compute

$$\text{RR}_{k,m,i} = 1 - \frac{|\text{NAI}_i^m|}{|\text{NAI}_i^{B0}|}, \quad (2)$$

and report the mean rr with bootstrap 95% CIs and Benjamini–Hochberg adjusted p -values.

Controlled-history slope (β_{norm}). For the conversation-history suite, we regress the model’s stage-2 estimate on the controlled anchor $a \in \{10, 30, 50, 70, 90\}$ and normalize the OLS slope: $\beta_{\text{norm}} = \beta \cdot 60/80$.

6 Mitigation strategies

We evaluate nine strategies organized into four categories:

Prompt-level. B1 (Ignore-Numbers): Prepends “Ignore any numbers that appear before the question. Base your answer only on the evidence provided.” B2 (Bias-Warning): Prepends “Anchoring bias causes estimates to be pulled toward irrelevant numbers. Adjust your reasoning to avoid this.” B3 (Chain-of-Thought): Replaces the answer footer with “Think step by step ...then give your final answer as a single integer.”

Structural. EFR (Evidence-First Repositioning): Moves the anchor block to immediately before the answer footer (anchor-last), so the model processes evidence before encountering the anchor. CxDP (Context-Distilled Prompting): Two-stage: first distill the context into a qualitative summary (no numbers), then estimate from the summary alone.

Post-hoc. B4 (Self-Consistency): Returns the median of 5 samples per condition. PLD (Post-hoc Linear Debiasing): Corrects the answer via $\hat{y} = y - \hat{\beta}(a - 50)$, where $\hat{\beta}$ is a per-model slope estimated from baseline data.

Model	Ext.	ICL	RAG	Tool	Hist.
Llama-3.2-1B	0.14	–	–	–	–
Llama-3.2-3B	0.35	–	–	–	–
Llama-3.1-8B ^H	0.28	–	–	–	–
GPT-4o-mini	0.14	−0.06	0.13	0.13	0.03
Claude-3.5-Haiku	0.09	−0.10	0.14	0.10	0.07
Llama-3.1-8B ^{OR}	0.41	−0.02	0.20	−0.02	0.06

Table 1: Baseline nai (B0) by model and anchor-delivery suite. “Ext.” = external prompt-embedded anchors. ^H = HuggingFace, ^{OR} = OpenRouter. “–” = suite not evaluated for this model/backend combination. Higher absolute values indicate stronger anchoring. All external-suite values are positive, indicating a pull toward the high anchor.

Strategy	1B	3B	8B ^H	8B ^{OR}	GPT-4o	Haiku
B0 (baseline)	0.14	0.35	0.28	0.41	0.14	0.09
B1 (ignore)	0.18	0.26	0.24	0.35	0.11	0.09
B2 (warning)	0.14	0.38	0.21	0.22	0.18	−0.04
B3 (CoT)	0.36	0.45	0.42	0.44	0.45	0.63
EFR	0.08	0.49	0.32	0.26	0.13	0.10
CxDP	–	–	–	0.25	0.38	0.18
B4 (median)	0.31	0.33	0.32	0.38	0.16	0.08
PLD	−0.16	0.16	0.11	0.28	0.04	0.02
AALC $\alpha=0.5$	0.14	0.21	0.24	–	–	–
AALC $\alpha=1.0$	−0.02	0.01	0.01	–	–	–
AALC $\alpha=1.5$	−0.05	−0.08	−0.09	–	–	–

Table 2: External-suite nai by mitigation strategy and model. AALC requires logit access and is only applicable to open-weight models. “–” = not evaluated. B3 (CoT) increases anchoring across all models. AALC at $\alpha=1.0$ brings nai to near zero for all three open-weight models.

Logit-level. AALC (Anchor-Aware Logit Correction): For open-weight models with logit access, we compute

$$\ell_{\text{corrected}} = \ell_a - \alpha \cdot (\ell_a - \ell_c), \quad (3)$$

where ℓ_a and ℓ_c are the anchored and control logits over the integer token vocabulary, and $\alpha \in \{0.5, 1.0, 1.5\}$ is a correction strength parameter.

7 Results

7.1 Baseline anchoring (nai under B0)

Table 1 reports the baseline nai across models and suites.

All models show positive nai on the external suite, confirming susceptibility to prompt-embedded anchors. Llama-3.1-8B (OpenRouter) exhibits the strongest effect (nai = 0.41), while Claude-3.5-Haiku shows the weakest (0.09). The ICL suite yields small or negative nai values, suggesting that anchoring via in-context demonstrations is less effective than direct prompt embedding—or that models partially “see through” biased demonstrations. RAG and tool suites show moderate anchoring (nai $\approx$ 0.10–0.20) for API models.

7.2 Mitigation effectiveness

Table 2 summarizes the nai under each mitigation on the external suite.

Three key findings emerge:

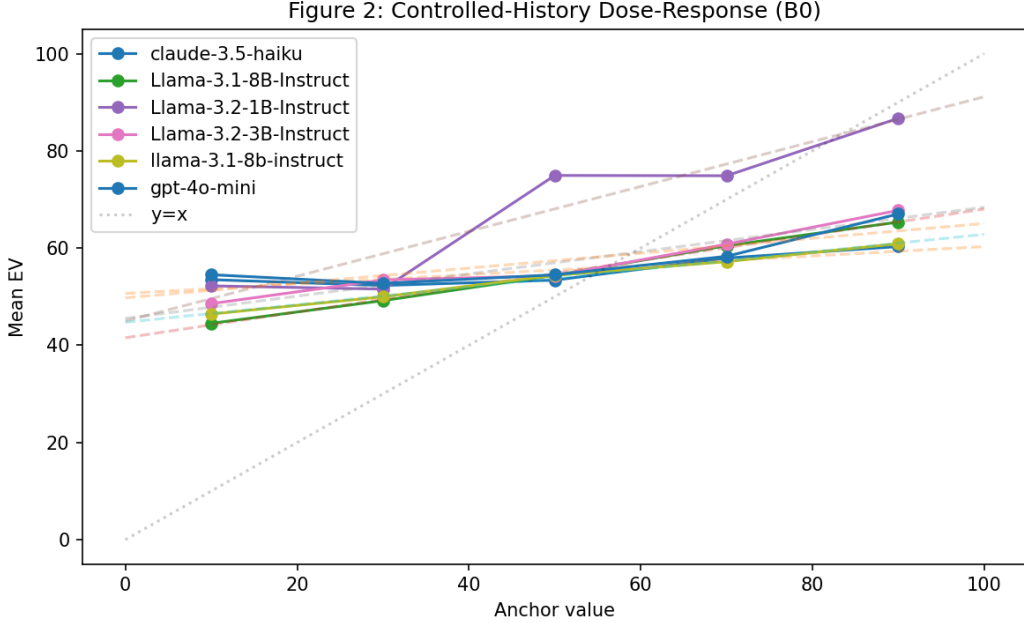


Figure 1: Dose-response: model EV vs. controlled anchor value in the conversation-history suite. A steeper slope indicates stronger anchoring by the model’s own prior answer.

Chain-of-thought amplifies anchoring. B3 (CoT) produces the highest nai values across all models (0.36–0.63), substantially exceeding baseline. Explicit step-by-step reasoning appears to give models more opportunity to incorporate the anchor into their reasoning chain.

Prompt-level mitigations are inconsistent. B1 and B2 reduce anchoring for some models (Claude-3.5-Haiku, Llama-3.1-8B via OpenRouter) but are ineffective or even counterproductive for others (Llama-3.2-1B, Llama-3.2-3B).

AALC is the most effective mitigation for open-weight models. At $\alpha=1.0$, AALC reduces external-suite nai from 0.14/0.35/0.28 to $-0.02/0.01/0.01$ for the 1B/3B/8B models, representing near-complete debiasing. At $\alpha=1.5$, mild over-correction is observed (negative nai).

7.3 Dose-response in controlled history

Figure 1 shows the relationship between the controlled anchor value $a \in \{10, 30, 50, 70, 90\}$ and the model’s stage-2 expected value (EV) for the conversation-history suite. A positive slope indicates that the model’s estimate is pulled toward the anchor.

7.4 AALC α sweep

Figure 2 shows how nai varies as a function of the AALC correction strength α across suites.

7.5 Stress tests

Table 3 reports nai under four stress-test conditions for the open-weight models (B0 baseline).

The 3B model is most sensitive to anchor position (T2: nai = 0.50) and verbal-format anchors (T3: 0.42), while the 8B model is most affected by RAG snippet ordering (T4:

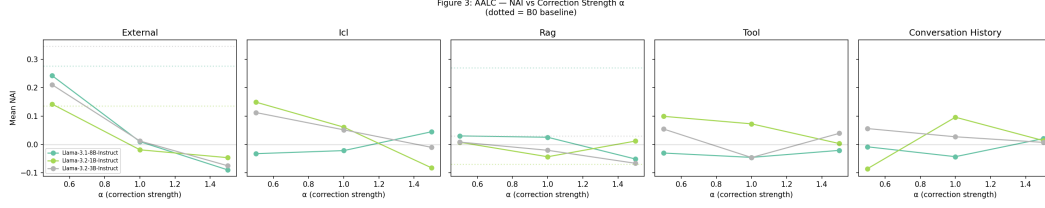


Figure 2: AALC α sweep: nai across suites for $\alpha \in \{0.5, 1.0, 1.5\}$. $\alpha=1.0$ consistently brings nai near zero; $\alpha=1.5$ over-corrects.

Test	Description	1B	3B	8B
T1	Extreme anchors (5/95)	0.24	0.23	0.04
T2	Anchor position	0.04	0.50	0.28
T3	Verbal-format anchor	-0.11	0.42	0.30
T4	RAG snippet order	-0.07	0.03	0.27

Table 3: Stress-test nai (B0) for open-weight models. $n=4$ items per test. T2 (position) and T3 (verbal format) reveal strong sensitivity for Llama-3.2-3B. T4 (RAG order) primarily affects the 8B model.

0.27). The 1B model shows moderate sensitivity to extreme anchors (T1: 0.24) but is relatively robust to position and format variations.

7.6 NAI vs. accuracy

Figure 3 visualizes the relationship between nai and task accuracy across models and mitigations.

8 Mechanistic interpretability of anchoring

We complement the behavioral analysis with a proof-of-concept mechanistic investigation that asks where in the transformer anchoring distortion arises and whether it can be localized to specific depth regions. All analyses use teacher-forced scoring (no sampling) on the external suite ($n=30$ items), applied to the three open-weight Llama models.

8.1 Methods

Anchor imprint. For each item we compute the integer output distribution $P(y)$ for $y \in \{0, \dots, 100\}$ under the control, low-anchor, and high-anchor prompts by extracting logits at the last token position and restricting to the single-token integer vocabulary. We measure the distributional shift via $\text{KL}(P_{\text{anchor}} \| P_{\text{control}})$, Total Variation Distance (TVD), Wasserstein-1 (W_1), and the expected-value shift $\Delta\text{EV} = \mathbb{E}[Y_{\text{anchor}}] - \mathbb{E}[Y_{\text{control}}]$. The EV shift is the distributional counterpart of the behavioral nai: $\text{NAI} = (\Delta\text{EV}_{\text{high}} - \Delta\text{EV}_{\text{low}})/60$.

Logit lens. At each transformer layer l , we apply the final layer norm and language-model head to the residual stream at the last token position, yielding a layer-wise integer distribution $P_l(y)$. We compute $\text{JS}(P_l^{\text{anchor}} \| P_l^{\text{control}})$ at every layer to identify the depth at which the anchor begins to distort the output distribution.

Residual-stream patching. We perform coarse causal attribution via activation patching:

1. Clean run: forward the control prompt, caching hidden states at every layer.
2. Patched run: forward the anchored prompt with hooks that replace the residual stream at the last token position for a contiguous bucket of layers with the cached clean states.

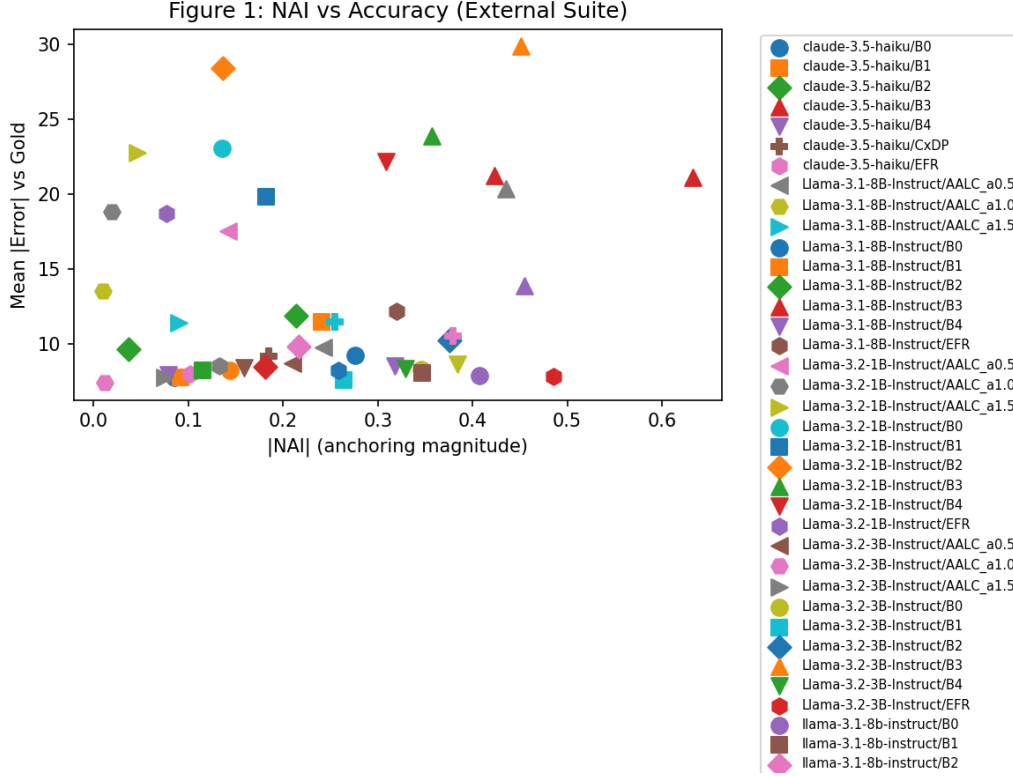


Figure 3: nai vs. accuracy across model-mitigation pairs. [TODO: Add interpretation once axis labels are verified.]

Model	KL		TVD		ΔEV	
	Low	High	Low	High	Low	High
Llama 1B	0.12	0.17	0.19	0.23	+4.8	+9.9
Llama 3B	0.62	1.70	0.36	0.66	-3.3	+20.6
Llama 8B	0.69	0.57	0.44	0.37	-8.4	+9.9

Table 4: Anchor imprint metrics (mean over $n=30$ external-suite items). $KL = KL(P_{\text{anchor}} \| P_{\text{control}})$; ΔEV = expected-value shift relative to control. The 3B model shows the largest high-anchor imprint (KL 1.70, ΔEV +20.6), consistent with its high behavioral nai (0.35).

184 Layers are divided into three equal-sized buckets (early, mid, late). The restoration metric
 185 quantifies how much of the anchor-induced divergence is removed:

$$\text{rest}(b) = 1 - \frac{D_{JS}(\hat{P}^b, P_{\text{ctrl}})}{D_{JS}(P_{\text{anch}}, P_{\text{ctrl}})}, \quad (4)$$

186 where $\hat{P}^b$ is the output distribution after patching bucket b . A restoration of 1 means the
 187 patch fully removes the anchoring distortion.

188 8.2 Anchor imprint results

189 Table 4 summarizes the distributional anchor imprint across models.

190 The 3B model shows the strongest distributional distortion under the high-anchor condi-
 191 tion (KL = 1.70, ΔEV = +20.6), consistent with its high behavioral nai of 0.35. The 8B
 192 model shows a more symmetric pattern (KL 0.69 low / 0.57 high) with EV shifts in oppo-

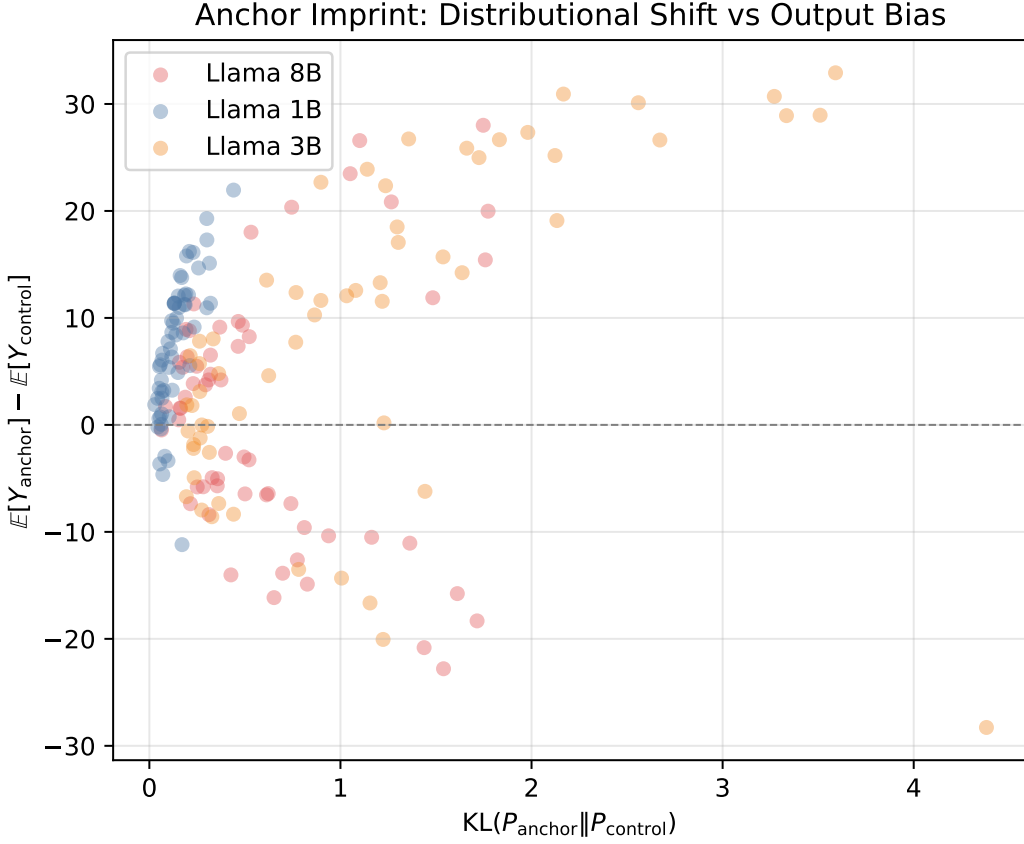


Figure 4: Anchor imprint scatter: KL divergence vs. EV shift for each item-condition pair. Each point is one item under either the low-anchor or high-anchor condition. Larger KL values correlate with larger absolute EV shifts, confirming that the anchor’s distributional effect drives the behavioral output bias.

193 site directions ($-8.4 / +9.9$), as expected from a model that is pulled toward whichever
 194 anchor is presented. The 1B model shows the weakest distributional shift overall, aligning
 195 with its lower behavioral nai (0.14). Figure 4 visualizes the per-item relationship between
 196 distributional divergence and output bias.

197 8.3 Logit-lens localization

198 Figure 5 shows the layer-wise JS divergence between anchored and control distributions.

199 Across all three models, the anchor-control divergence remains near zero in early layers
 200 and rises sharply in the final third of the network. This “late-onset” pattern suggests that
 201 the anchor does not corrupt early representations; rather, the distortion is introduced (or
 202 amplified) in the later layers that are most directly responsible for mapping representations
 203 to output tokens.

204 8.4 Patching localization

205 Table 5 and Figure 6 report the restoration achieved by patching each layer bucket.

206 The causal attribution results are striking in their consistency: patching the late third of
 207 layers restores $\geq 98\%$ of the anchoring divergence across all three model sizes. Early-layer
 208 patching has negligible effect ($\leq 7\%$). Mid-layer restoration is moderate for the 3B and 8B

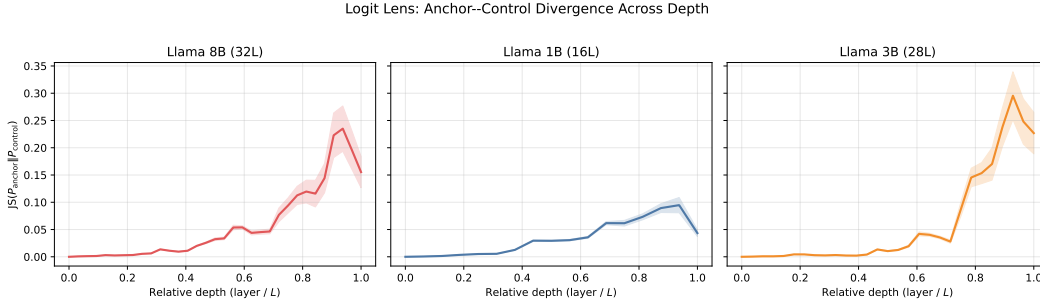


Figure 5: Logit lens: JS divergence between anchored and control integer distributions as a function of relative depth (layer l / total layers L). Shaded regions show 95% CIs across items. In all three models, divergence is near zero in early layers and rises sharply in the final third, indicating that the anchor distortion is concentrated in late layers.

Model	Early		Mid		Late	
	Mean	Std	Mean	Std	Mean	Std
Llama 1B	0.07	0.14	-0.16	0.67	0.98	0.02
Llama 3B	0.00	0.04	0.43	0.50	0.99	0.01
Llama 8B	0.04	0.12	0.38	0.26	0.99	0.01

Table 5: Residual-stream patching restoration by layer bucket. Replacing the late-third hidden states with clean (control) states restores $\geq 98\%$ of the anchoring divergence across all models. Early layers contribute minimally ($\leq 7\%$). Mid-layer restoration is variable, especially for the 1B model.

models (0.43 and 0.38) but noisy for the 1B model (-0.16 ± 0.67), suggesting that the 1B model’s anchoring mechanism is more distributed across the mid-late boundary.

These results are consistent with the logit-lens findings and jointly suggest that anchoring distortion in these models is a late-layer phenomenon: the anchor token is processed through the full context, but the actual shift in the integer output distribution occurs predominantly in the layers closest to the output head.

8.5 Limitations of the mechanistic analysis

External suite only. The patching design requires paired control/anchor prompts of similar structure, which is cleanly available only for the external suite. ICL, RAG, and tool suites involve structural prompt differences that confound patching.

Coarse granularity. We partition layers into three buckets for robustness across model sizes. This cannot pinpoint individual layers, attention heads, or MLP sublayers. Finer-grained causal tracing (e.g., per-layer, per-head) is left for future work.

Single-token approximation. The logit-lens and patching analyses score only single-token integers. Multi-token answers (e.g., “100”) are handled correctly in the imprint analysis but not in the layer-wise diagnostics.

No attention-head attribution. We do not perform head-level ablation or probing. Identifying specific “anchor heads” or circuits requires more targeted interventions. [TODO: Add attention-head ablation or probing in future work.]

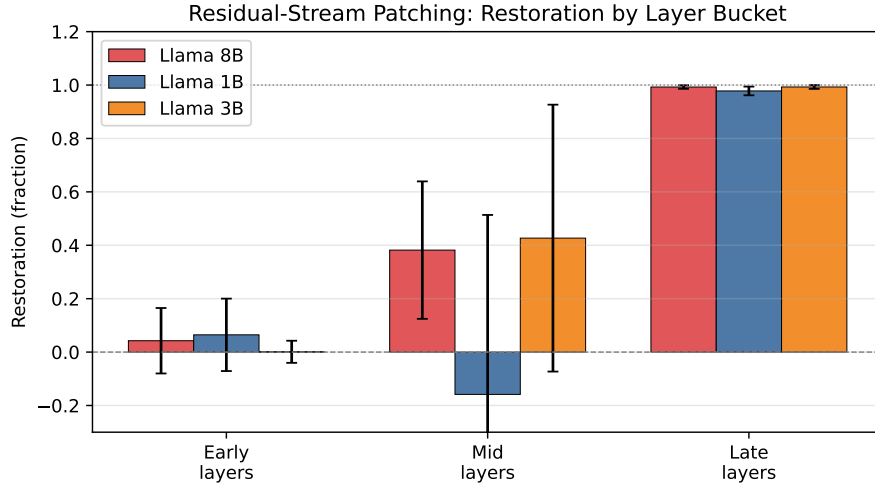


Figure 6: Patching restoration by layer bucket across models. Error bars show standard deviation across items and conditions. Late-layer patching consistently restores nearly all of the anchoring distortion.

228 9 Limitations and next steps

229 PoC scope. This work is a proof of concept. The benchmark contains 108 items—sufficient
 230 for detecting large effects but too small for fine-grained per-domain analysis. We plan to
 231 scale to ≥ 500 items with broader domain coverage.

232 Model coverage. We evaluate six model configurations across three families. Larger models
 233 ($\geq 70B$), reasoning-specialized models, and additional API providers remain untested.

234 Single language. All items are in English. Anchoring susceptibility may vary across lan-
 235 guages.

236 Synthetic anchors. Our anchors are constructed rather than naturally occurring. Real-
 237 world anchoring (e.g., from web retrieval or user-provided context) may differ in strength
 238 and mechanism.

239 AALC limitations. AALC requires logit access (open-weight models only) and a paired
 240 control prompt, making it inapplicable to black-box APIs. The correction strength α must
 241 be tuned per deployment.

242 Next steps. We plan to: (1) expand the benchmark to 500+ items with adversarial anchor
 243 placements; (2) evaluate on models $\geq 70B$ and reasoning-specialized models; (3) extend to
 244 live RAG and tool-use pipelines; (4) develop AALC variants that do not require a paired
 245 control prompt; (5) investigate the mechanism by which CoT amplifies anchoring.

246 Author contributions

247 [TODO: Fill in author contributions before submission.]

248 Acknowledgments

249 [TODO: Add acknowledgments and funding information.]

Ethics statement

This work studies a failure mode of LLMs that could lead to systematically biased outputs in high-stakes applications. Our benchmark uses synthetic scenarios in health, finance, and operations domains; no real patient, financial, or operational data is used. We release the benchmark to enable further research on LLM robustness.

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A Reproducibility

Dataset. The benchmark is generated by poc_dataset/generate_poc_dataset.py (v0.1, 60 items) and extended by poc_dataset/extend_poc_dataset_v0_2.py (v0.2, 108 items). The v0.2 dataset is stored at poc_dataset/poc_v0.2.jsonl.

Inference. Open-weight models are run via runner/run_hf.py (HuggingFace Transformers, bfloat16, device_map=auto). API models are run via runner/run_openrouter.py (OpenRouter).

Mitigations. Mitigation experiments use mitigation/run_mitigation_hf.py (local), mitigation/run_mitigation_async.py (API), and mitigation/run_aalc.py (AALC).

294 Evaluation. Metrics are computed by mitigation/eval_mitigations.py and mitiga-
295 tion/generate_results.py.

296 Configuration. Inference settings: temperature = 0.7, top- p = 1.0, max_tokens = 8 (128
297 for B3), n = 5 samples per item. AALC tested at $\alpha \in \{0.5, 1.0, 1.5\}$. All configs are stored
298 in mitigation/config.yaml and runner/config.models.yaml.

299 Stress tests. Generated by mitigation/generate_stress.py; stored at
300 poc_dataset/stress_test.jsonl.

301 [TODO: Add hardware details (GPU type, total runtime).] [TODO: Add random seeds
302 used for sampling.]